Higher-Dimensional Consistency of Geometry and Modular Flow: Small-Ball Modular Data and Linearized Gravity

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Abstract

Paper I established that, in 1+1 dimensions, emergent geometry and modular flow are not independently deformable. The higher-dimensional sequel cannot proceed by simply replacing the one-parameter cross-ratio family with a higher-dimensional analogue. For ball-shaped regions in $d+1$ dimensions, the relevant non-gauge datum is not an individual nested configuration but the full family of modular Hamiltonians attached to all balls and all local rest frames. The new question is therefore one of consistency: when can this multi-directional family of modular data be explained by a single geometry?

We study this question in the CHM ball framework for conformal nets with finite-dimensional conformal covariance, in the linearized regime. First, we show that the small-ball limit of the modular energy uniquely recovers the local stress tensor from the all-directions family of modular data. Second, for a local geometric functional satisfying linearity, locality, covariance, gauge invariance, and the correct scaling behavior, we additionally assume that its leading coefficient is a local symmetric rank-2 tensor of exactly two-derivative order and is divergence-free. Under those independently supplied premises, the linearized Lovelock classification identifies that coefficient with the linearized Einstein tensor up to normalization. Matching the modular and geometric small-ball expansions then yields the linearized Einstein equation within the stated domain. Thus, in higher dimensions, inseparability does not merely lock geometry and modular flow together; the consistency of that locking across directions takes the form of linearized gravitational constraint equations.

The resulting theorem is conditional but precise. It does not attempt full generality, nonlinear dynamics, or arbitrary Type III_1 nets. Its proof domain is restricted to CHM ball data in conformally covariant theories, in the linearized regime. Within that domain, however, the conclusion is sharp: the linearized Einstein equation appears as the consistency condition required for a single geometry to remain compatible with the full modular data family.

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1 Introduction

Paper I[1] showed that, in $1+1$ dimensions, the standard hard-problem presupposition of independent describability fails because geometry and modular flow cannot be independently deformed. That result depended on a one-parameter datum: the cross-ratio family of double intervals. In higher dimensions, the mathematical problem changes in a decisive way. There is no longer a single distinguished scalar modulus whose variation carries the entire scale information. The higher-dimensional sequel must therefore identify a different object.

The central claim of the present paper is that the correct object is the full family of ball modular Hamiltonians, indexed not only by center and radius but also by local timelike direction. Once this is recognized, the problem becomes a consistency problem rather than a parametrization problem. A single ball gives only one local modular datum. A family of balls in many directions gives many such data. The question is whether they can all be explained by one and the same geometry.

Our conditional answer is that, in the CHM ball framework and at linear order, consistency yields an Einstein coefficient equation once a local, two-derivative, divergence-free geometric coefficient is supplied. The CHM data alone recover the local stress tensor; they do not select the geometric operator.

The proof strategy is intentionally narrow. We do not attempt a nonlinear theorem, a fully general Type III_1 result, or an "if and only if" in complete abstractity. Instead we isolate a tractable domain in which the statement can be made cleanly: conformal nets with finite-dimensional conformal covariance, CHM modular Hamiltonians for balls, the vacuum or near-vacuum linearized regime, and a local geometric functional satisfying a small list of natural structural axioms. Within that domain, the theorem is short. The real work lies in identifying the correct small-ball expansions on both the modular and geometric sides.

The paper is organized in the reverse order of discovery. Section 2 defines the higher-dimensional modular datum and explains why the natural $1+1$-dimensional cross-ratio strategy has no direct higher-dimensional analogue. Section 3 formulates the small-ball consistency condition that replaces assembly in higher dimensions. Section 4 states and proves the main theorem. Section 5 records the free-field worked example that motivated the final theorem statements. Section 6 discusses the converse direction and the limits of the argument. Section 7 places the result in the broader Consciousness Trilogy, and Section 8 lists the natural open problems.

Remark 1.1 (Relation to Paper I). Paper II is mathematically self-contained. It uses Paper I only for motivation and context: the conceptual shift from "independent describability" to "independent deformability," and the observation that in $1+1$ dimensions the cross-ratio family carries the full scale information. No theorem from Paper I is invoked as a premise in any proof of Paper II. The CHM ball framework, the geometric axioms (G1)–(G5), and the small-ball consistency condition (MC) are all defined and used within the present paper.

2 Multi-Directional Modular Data in Higher Dimensions

2.1 From Cross-Ratio to Ball Families

Paper I[1] relied on the fact that, in $1+1$ dimensions, the correct non-gauge local modulus is the cross-ratio of a double-interval configuration. That datum is one-dimensional, and the resulting assembly problem is trivial. There is only one parameter direction, so there is no compatibility problem between different directional slices.

In higher dimensions, the naive analogue fails. A single ball is a conformal gauge orbit. Even a nested pair of balls does not produce a nontrivial scalar modulus in the way that a double interval does in $1+1$ dimensions. The correct higher-dimensional object therefore cannot be "a better cross-ratio." The no-go is structural: the relevant conformal degrees of freedom are not carried by one isolated nested configuration.

What replaces the cross-ratio is the family of all ball modular data. The local region is now a ball

$$B(\mathbf{x}_0, R) \subset \mathbb{R}^d,$$

or, more invariantly, a small causal diamond with center x_0, radius R, and rest-frame normal u^μ. Each such region gives one modular Hamiltonian. The higher-dimensional datum is not one scalar parameter but a family

$$\{K_{B,u}\}_{(B,u)\in\mathcal{B}},$$

where $\mathcal{B}$ denotes the set of all sufficiently small balls in all local rest frames.

2.2 CHM Ball Hamiltonians

In the vacuum of a conformal theory, the modular Hamiltonian of a ball is local and universal. In a rest frame adapted to the ball, the Casini–Huerta–Myers[2] formula gives

$$K_{B,u} = 2\pi \int_{B_u} d^d x \, \frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R} \, T_{\mu\nu}(x)\, u^\mu u^\nu.$$

The importance of this formula for the present paper is twofold. First, it supplies an explicit local kernel. Second, it shows that what the ball modular Hamiltonian probes is not an arbitrary operator but the rest-frame energy density $T_{uu} := T_{\mu\nu}u^\mu u^\nu$.

This already reveals the first higher-dimensional subtlety. If one restricts to a single fixed time slice, then the family of all balls only probes T_{00}. That is enough to recover the Hamiltonian constraint source, but not enough to recover the full stress tensor. To reconstruct the full tensorial source, the family must vary not only in center and radius but also in local timelike direction. This is why "multi-directional modular datum" is not rhetorical excess but a necessary mathematical strengthening.

2.3 The Canonical Datum

We therefore take the higher-dimensional canonical modular datum to be the full ball family

$$K_{B,u} := -\ln \Delta_{\mathcal{A}(B_u),\Omega},$$

understood in the CHM class where the modular Hamiltonian admits the local stress-tensor representation above. This datum is canonical in exactly the sense needed for the sequel to Paper I[1]: it is the minimal non-gauge family rich enough to encode the directional dependence that becomes nontrivial in dimensions higher than $1+1$.

The conceptual shift is worth making explicit. In Paper I[1], geometry and modular flow were locked by a one-parameter scale datum. In the present paper, they are locked by the requirement that an entire family of local modular energies be compatible with one geometry at once. What becomes nontrivial in higher dimensions is therefore not "finding the right modulus," but enforcing coherence across a family of moduli.

3 The Small-Ball Consistency Condition

3.1 Why Small Balls First

The strongest possible version of Paper II would formulate a global consistency condition on the modular data of all balls and derive the linearized Einstein equation from that global condition. We do not begin there. The small-ball regime already contains the local dynamical content, while avoiding the additional integrability problems of a global ball-space formulation. If the thesis fails even in the small-ball limit, the global version has no chance. If it succeeds there, the globalization problem becomes a genuine strengthening rather than a prerequisite.

3.2 Modular Small-Ball Expansion

Let $\delta\omega$ be a linearized perturbation of the vacuum, and assume the local stress tensor is smooth in a neighborhood of a point x_0. Then the CHM kernel gives an immediate small-ball asymptotic:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}),$$

where $V_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit d-ball. The key point is not the coefficient but the structural content: the leading small-ball term is the quadratic form of the local stress tensor evaluated on the timelike direction u^μ.

Because the family is taken over all timelike u^μ, the leading term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$. This is the first half of the main theorem. The all-directions small-ball modular datum is already strong enough to recover the complete local matter source.

3.3 Geometric Small-Ball Functional

The geometric side is intentionally abstract. We do not assume a particular holographic dictionary, RT formula[6], or bulk construction. Instead we assume that the geometry supplies, for each small ball and each direction, a linearized functional

$$\delta\mathcal{G}_{B,u}[h],$$

depending on the metric perturbation $h_{\mu\nu}$, subject to five structural requirements:

1. linearity in h;
2. locality in the small-ball limit;
3. covariance under local Lorentz transformations;
4. invariance under linearized gauge transformations;
5. the same leading scaling R^{d+1} as the modular side.

These assumptions are not arbitrary. They are exactly what is needed for a small-ball geometric quantity to be comparable to the CHM modular energy without smuggling in additional dynamics by hand.

3.4 Consistency

The higher-dimensional consistency condition can now be stated in its minimal form.

Small-ball modular consistency (MC). For all sufficiently small balls and all local timelike directions u^μ, the modular small-ball functional and the geometric small-ball functional agree:

$$\delta\mathcal{G}_{B,u}[h] = \delta\langle K_{B,u}\rangle.$$

This is the higher-dimensional replacement for the trivial assembly of Paper I[1]. In $1+1$ dimensions, there is only one parameter direction, so the scale datum needs no consistency check beyond existence. In higher dimensions, the requirement that all directional slices fit one geometry is itself the dynamical content.

4 Conditional Einstein Coefficient Matching

4.1 Modular Extraction Proposition

The modular side is summarized by the following statement.

Proposition 4.1. *In the CHM ball framework, for a linearized perturbation of the vacuum with a smooth local stress tensor, the all-directions small-ball modular datum has leading expansion*

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}).$$

The proof is the small-ball Taylor expansion of the CHM kernel, together with the symmetry of the ball that kills the linear term. The resulting leading coefficient is universal and independent of the detailed model.

Corollary 4.2. *The all-directions leading small-ball term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$ at each point.*

This is immediate because a symmetric rank-2 tensor is uniquely fixed by its quadratic form on all timelike directions.

4.2 Missing Premises for Geometric Identification

The former claim that (G1)–(G5) alone select the linearized Einstein tensor is withdrawn. Those conditions do not restrict derivative order, and even at two derivatives the relevant coefficient must separately be required to be transverse. The following premises are therefore not consequences of (G1)–(G5):

(G-der2) The leading coefficient of $\delta\mathcal{G}_{B,u}[h]$ is a local symmetric rank-2 tensor, linear in h, of exactly two derivative order.

(G-div) That coefficient is divergence-free on the Minkowski background.

When (G-der2), (G-div), linearized gauge invariance, and Lorentz covariance are supplied, the standard linearized Lovelock classification identifies the coefficient with $\alpha_d G^{\mathrm{lin}}_{\mu\nu}$ up to normalization [5]. This is a conditional coefficient identification, not a uniqueness result derived from (G1)–(G5).

4.3 Conditional Coefficient Matching

With the missing geometric premises displayed, coefficient matching is immediate.

Theorem 4.3 (Conditional Einstein coefficient matching)**.** *Assume:*

1. *CHM modular Hamiltonians for all sufficiently small balls;*
2. *conformal covariance of the net (finite-dimensional conformal group* $\mathrm{SO}(d,2)$*);*
3. *linearization around a Minkowski vacuum background;*
4. *existence of a geometric functional satisfying the small-ball axioms (G1)–(G5) above;*
5. *the additional premises (G-der2) and (G-div);*
6. *small-ball modular consistency (MC).*

Then

$$G^{\mathrm{lin}}_{\mu\nu} = \kappa_d\, \delta\langle T_{\mu\nu}\rangle,$$

with $\kappa_d = \frac{2\pi V_d}{\alpha_d(d+2)}$.

The proof first uses the explicitly supplied (G-der2) and (G-div), together with gauge invariance and covariance, to invoke the linearized Lovelock classification. It then compares the resulting $G^{\mathrm{lin}}_{\mu\nu}u^\mu u^\nu$ coefficient with the CHM stress-tensor coefficient. Equality for all timelike u^μ forces equality of the tensors themselves.

Remark 4.4 (Conditionality and consistency). Assumptions (4)–(5) are the most significant: neither the geometric functional nor its two-derivative transverse leading coefficient is derived from the algebraic data. The theorem says that *if* such a functional exists and agrees with the modular side for all sufficiently small balls, *then* its leading local content is forced to be Einsteinian. The internal construction of $\delta\mathcal{G}_{B,u}$ from modular data alone remains the most significant open problem (Section 8.3).

The matching is also dynamically consistent: the linearized Einstein tensor satisfies the Bianchi identity $\partial^\mu G^{\mathrm{lin}}_{\mu\nu} = 0$, and the stress tensor satisfies the conservation law $\partial^\mu\delta\langle T_{\mu\nu}\rangle = 0$. Both sides of the equation are independently divergence-free.

4.4 Interpretation

This theorem is the higher-dimensional analogue of the inseparability theorem of Paper I[1], but with one crucial new feature. In $1+1$ dimensions, inseparability rules out independent deformation. In higher dimensions, the demand that the inseparable structures remain mutually consistent across all directions forces the geometry to obey linearized gravitational dynamics.

That is the sense in which gravity enters the trilogy. It is not introduced as an additional law attached from outside the modular-geometric relation. It appears as the consistency condition required for that relation to survive directional proliferation.

5 Worked Example: Free Conformal Scalar Field in $d+1$ Dimensions

This section verifies the main theorem in the simplest higher-dimensional model where all ingredients are explicit: the free massless conformal scalar field in $d+1$ dimensions. The verification has three parts: the stress tensor and its perturbation, the small-ball modular expansion, and the consistency check against the linearized Einstein equation.

5.1 The Model

Let ϕ be a free massless scalar field in $d+1$ Minkowski spacetime, with action

$$S = -\frac{1}{2}\int d^{d+1}x\,(\partial\phi)^2.$$

The stress-energy tensor is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}(\partial\phi)^2,$$

with the conformal improvement term understood when $d > 1$. In the vacuum state $|\Omega\rangle$, $\langle T_{\mu\nu}\rangle = 0$.

5.2 Perturbation: Classical Source

For the purpose of verifying the small-ball expansion, we do not need the full machinery of coherent states. It suffices to take any smooth, conserved classical stress tensor as the source. Let ϕ_{cl} be a smooth solution of the free wave equation $\Box\phi_{\text{cl}} = 0$, and define

$$\delta\langle T_{\mu\nu}(\mathbf{x})\rangle := \partial_\mu\phi_{\text{cl}}\,\partial_\nu\phi_{\text{cl}} - \frac{1}{2}\eta_{\mu\nu}(\partial\phi_{\text{cl}})^2,$$

with the conformal improvement understood when $d > 1$. This is the classical stress tensor of the on-shell configuration ϕ_{cl}. It is smooth, symmetric, and satisfies the conservation law

$$\partial^\mu\delta\langle T_{\mu\nu}\rangle = 0$$

as a direct consequence of $\Box\phi_{\text{cl}} = 0$.

The "linearized regime" of the theorem refers to linearization in the metric perturbation $h_{\mu\nu}$, not in the matter amplitude. The classical source $\delta\langle T_{\mu\nu}\rangle$ is a fixed, smooth tensor field that enters the right-hand side of the linearized Einstein equation. No conflict arises between the quadratic nature of the stress tensor in ϕ_{cl} and the linearized-gravity framework.

5.3 Modular Side: Small-Ball Expansion

For a ball $B(\mathbf{x}_0, R)$ in the rest frame with timelike normal $u^\mu = (1, \mathbf{0})$, the CHM formula gives

$$\delta\langle K_B\rangle = 2\pi\int_{|\mathbf{x}-\mathbf{x}_0|<R} d^dx\,\frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R}\,\delta\langle T_{00}(\mathbf{x})\rangle.$$

Expanding $\delta\langle T_{00}(\mathbf{x})\rangle$ around $\mathbf{x}_0$:

$$\delta\langle T_{00}(\mathbf{x})\rangle = \delta\langle T_{00}(\mathbf{x}_0)\rangle + \partial_i\delta\langle T_{00}\rangle(\mathbf{x}_0)(x^i - x_0^i) + O(|\mathbf{x}-\mathbf{x}_0|^2).$$

The linear term integrates to zero by the spherical symmetry of the ball. The leading contribution is therefore

$$\delta\langle K_B\rangle = 2\pi\,\delta\langle T_{00}(\mathbf{x}_0)\rangle \int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} + O(R^{d+3}).$$

The radial integral evaluates to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{\Omega_{d-1}}{2R}\int_0^R r^{d-1}(R^2-r^2)\,dr = \frac{\Omega_{d-1}}{2R}\left(\frac{R^{d+2}}{d}-\frac{R^{d+2}}{d+2}\right) = \frac{\Omega_{d-1}\,R^{d+1}}{d(d+2)},$$

where $\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the surface area of the unit $(d-1)$-sphere. Since $V_d = \frac{\Omega_{d-1}}{d}$, this simplifies to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{V_d\,R^{d+1}}{d+2}.$$

Therefore

$$\delta\langle K_B\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{00}(\mathbf{x}_0)\rangle + O(R^{d+3}).$$

For a boosted ball with timelike normal u^μ, the same computation in the adapted frame gives

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\,u^\mu u^\nu + O(R^{d+3}),$$

confirming Proposition 4.1 in the free-field model.

5.4 Full Tensor Recovery

In the free-field classical source, $\delta\langle T_{\mu\nu}\rangle$ is a smooth symmetric tensor field. By Corollary 4.2, the all-directions small-ball data uniquely determine this tensor at each point. In the present model, this can be verified directly: the quadratic form $u^\mu u^\nu\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ for all timelike u determines $\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ because the timelike cone spans the full symmetric tensor space.

5.5 Conservation Law

The classical stress tensor satisfies the conservation law

$$\partial^\mu\delta\langle T_{\mu\nu}\rangle = 0$$

as established in Section 5.2 from the on-shell condition $\Box\phi_{\text{cl}} = 0$. This is the necessary compatibility condition for the comparison with the linearized Einstein tensor, which satisfies the Bianchi identity $\partial^\mu G^{\text{lin}}_{\mu\nu} = 0$. In the free-field model, both sides of the would-be Einstein equation are independently divergence-free, so the matching is dynamically consistent and not merely algebraic.

5.6 Consistency Check

Suppose a linearized metric perturbation $h_{\mu\nu}$ is sourced by the free-field stress tensor via the semiclassical Einstein equation

$$G^{\text{lin}}_{\mu\nu}[h] = 8\pi G_N\,\delta\langle T_{\mu\nu}\rangle.$$

If one additionally supplies a geometric small-ball functional whose leading coefficient has the (G-der2) and (G-div) form used in Theorem 4.3, then it has leading term

$$\delta\mathcal{G}_{B,u}[h] = \alpha_d\, R^{d+1}\, G^{\mathrm{lin}}_{\mu\nu}(\mathbf{x}_0)\, u^\mu u^\nu + O(R^{d+3}) = \alpha_d \cdot 8\pi G_N\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

Comparing with the modular side:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

The consistency condition (MC) requires these to be equal. Matching coefficients:

$$\alpha_d \cdot 8\pi G_N = \frac{2\pi V_d}{d+2},$$

which fixes the gravitational coupling in terms of the geometric normalization constant α_d and the spatial dimension d. This is not a derivation of G_N from first principles — the constant α_d is part of the geometric functional's normalization — but it shows that the coefficient comparison is compatible with the linearized Einstein equation: if such an equation holds, the modular and geometric small-ball data match at leading order.

5.7 What the Example Does and Does Not Show

The free-field example verifies three things:

1. The small-ball modular expansion (Proposition 4.1) is not a formal artifact. It is realized with explicit, computable coefficients in a standard QFT model.

2. The conservation law on the modular side matches the Bianchi identity on the geometric side, so the comparison is dynamically consistent.

3. The coefficient comparison is compatible with a semiclassical Einstein interpretation: if a metric perturbation satisfying the linearized Einstein equation exists, then the consistency condition (MC) is satisfied at leading order.

The example does not construct the geometric functional and therefore does not establish the existence premise of the conditional theorem. It assumes the semiclassical Einstein equation and checks compatibility of that assumption with the CHM modular coefficient. It does not show that (MC) is the unique route to the Einstein equation, nor does it address the converse direction.

6 Converse, Limits, and Relation to Faulkner–Jacobson

6.1 Why the Theorem Is Not an Equivalence

Theorem 4.3 is a conditional coefficient-matching theorem. It begins with the small-ball modular family and additionally assumes a single geometric explanation with the two-derivative and transverse premises (G-der2) and (G-div). Under those supplied premises it concludes that the leading coefficients obey the linearized Einstein equation. A reverse implication would require substantially more structure. One would need a faithful reconstruction map

$$h_{\mu\nu} \longmapsto \{\delta\mathcal{G}_{B,u}[h]\}_{(B,u)}$$

defined for all sufficiently small balls, with the correct normalization and covariance properties, and with agreement not only at leading order but at the level of the full ball-dependent functional. The linearized Einstein equation by itself does not provide such a map.

This distinction is not pedantic. The coefficient comparison in Section 4 identifies the leading local tensorial term,

$$G^{\mathrm{lin}}_{\mu\nu}(x_0) \propto \delta\langle T_{\mu\nu}(x_0)\rangle,$$

because equality for all timelike u^μ forces equality of the corresponding quadratic forms. But that argument is deliberately local and asymptotic. It does not control the $O(R^{d+3})$ remainder, it does not determine the full dependence of $\delta\mathcal{G}_{B,u}[h]$ on the ball parameters, and it does not show that every solution of the linearized Einstein equation arises from a geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The converse direction therefore belongs to a stronger future theorem, not to the present one.

6.2 Precise Relation to the Faulkner–Jacobson Line

The closest existing line of work runs from the CHM ball modular Hamiltonian through the entanglement first law to the Faulkner[3]–Jacobson[4] derivations of linearized gravity (see also [8]). The overlap is genuine and should be stated without exaggeration. In both settings, ball-shaped regions are essential, the CHM kernel supplies the universal local weighting, and the comparison across sufficiently small balls in all directions forces a rank-2 tensor equation whose leading local content is Einsteinian. At the level of small-ball tensor algebra, both routes exploit the same rigid fact: on a Minkowski background, the only local, symmetric, gauge-invariant, two-derivative tensor linear in $h_{\mu\nu}$ is the linearized Einstein tensor up to normalization [5].

The difference lies in what is taken as primitive. Faulkner-type arguments begin from an entropy variation on the geometric side, typically mediated by holographic ingredients such as the RT formula[6] and, in the quantum-corrected setting, JLMS[7]. The present paper begins instead from the family of modular Hamiltonians itself and asks under which supplied geometric conditions one geometry can explain that family. The geometric side is therefore left abstract: we do not assume a bulk dual, a surface-area formula, or an entanglement-entropy interpretation. We do assume a small-ball geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The result is thus explicit about where the geometric input enters.

It is therefore best to regard the two routes as overlapping but not identical. If one instantiates $\delta\mathcal{G}_{B,u}[h]$ by an entropy or area variation of the appropriate kind, then the present framework reproduces the local coefficient comparison of the Faulkner–Jacobson argument. Without such an instantiation, Theorem 4.3 remains a conditional consistency statement rather than an internal derivation of gravity.

6.3 Limits of the Present Result

The domain of validity should also be kept explicit. First, the modular side is restricted to the CHM ball class in conformally covariant theories, so the theorem does not apply to arbitrary Type III_1 nets. Second, the argument is local and small-ball. It extracts the leading tensorial content near each point, but it does not yet provide a global all-balls integrability theorem. Third, the analysis is linearized around a Minkowski vacuum background. Extension to other maximally symmetric backgrounds (de Sitter, anti-de Sitter) would require replacing $G^{\mathrm{lin}}_{\mu\nu}$ by the appropriate background-relative linearized Einstein operator; this is a natural but nontrivial generalization. Nothing in the present paper proves the full nonlinear Einstein equation.

Fourth, the geometric functional and the premises (G-der2) and (G-div) are axiomatized rather than internally derived from the algebraic data. This is the main remaining mathematical burden. The theorem shows that once these inputs exist and agree with the modular side for all sufficiently small balls, their leading coefficients match in Einstein form. It does not explain why an admissible algebraic model should supply those inputs. Globalization, nonlinear order, and internal construction are therefore genuine next problems rather than hidden assumptions.

7 Broader Physical and Conceptual Implications

Paper I[1] gives a conditional algebraic inseparability result. The present paper asks what an explicitly restricted higher-dimensional coefficient comparison adds to that programme.

In $1+1$ dimensions, the conditional inseparability statement is a rigidity result. In higher dimensions, if a geometric functional also satisfies (G-der2) and (G-div), all-direction consistency gives a linearized Einstein coefficient equation. The dynamics is therefore identified only inside this supplied geometric regime.

This has three consequences for the trilogy as a whole.

First, it isolates the extra premises required to obtain an Einstein coefficient. The equation is not derived from modular consistency alone: the two-derivative and divergence-free geometric structure is supplied. Within that regime, all-direction consistency fixes the coefficient matching.

Second, it gives a conditional higher-dimensional extension of Paper I's programme. The gravitational coefficient equation follows only within the stated CHM, linearized, and supplied-geometric regime; it is not a general dynamical consequence of modular data.

Third, the $O(R^{d+3})$ remainder remains uncontrolled. This paper does not construct coefficient directions, a Fisher map, or a self-description budget from that remainder. Accordingly it supplies no remainder-to-self-description bridge for Paper III.

8 Open Problems

8.1 Globalization

The present theorem is small-ball and local. It extracts the leading tensorial content near each point but does not provide a global all-balls integrability theorem. A global version would require controlling the full ball-dependent functional $\delta\mathcal{G}_{B,u}[h]$, not just its small-ball asymptotic. This is a genuine strengthening that would bring the result closer to the Faulkner[3]–Jacobson[4] line, where the entanglement first law is applied to all balls simultaneously.

8.2 Nonlinear Order

The $O(R^{d+3})$ and higher terms in the small-ball expansion may encode nonlinear corrections to the Einstein equation. In the Jacobson[4] thermodynamic derivation, the passage from linearized to full nonlinear Einstein equations requires an additional equilibrium or maximal-entropy assumption. Whether a similar upgrade is available in the present modular-consistency framework is an open question. This is the natural bridge to a later gravity paper, not something to force into the present one.

8.3 Internal Construction of the Geometric Functional

At present, the geometric small-ball functional $\delta\mathcal{G}_{B,u}[h]$, (G-der2), and (G-div) are axiomatized, not constructed from purely algebraic data. The theorem identifies an Einstein coefficient only after those inputs are supplied. Constructing the functional and deriving its derivative order and transversality from modular data would be required to remove this conditionality.

8.4 No Current Coefficient-Tower Bridge

The higher-order remainder is only an asymptotic error term in the present analysis. No independent coefficient directions or Fisher map are constructed here, so it cannot be passed to Paper III as a physical control tower or quantitative self-description deficit. Establishing such a bridge would require new definitions and estimates beyond the current small-ball result.

8.5 Relation to Entanglement Equilibrium

Jacobson's[4] entanglement equilibrium approach derives the full nonlinear Einstein equation from the assumption that the vacuum maximizes entanglement entropy for small causal diamonds. The present framework does not assume entropy maximization. Whether the modular consistency condition (MC) implies, or is implied by, an entanglement equilibrium condition is an unexplored connection that could unify the two approaches.

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